

SHLOK CHANNAWAR

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EDUCATION

The Pennsylvania State University
Bachelor of Science in Applied Data Science

State College, PA
December 2027

TECHNICAL SKILLS

Languages: Python (PyTorch, TransformerLens, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib), R, SQL
ML/DL: Neuronpedia Hugging Face (Transformers, Datasets), Weights & Biases
Tools: Git, Jupyter, VS Code, Lambda Cloud, Google Colab

RESEARCH

- Look Before You Steer: Geometry Predicts SAE Feature Steerability** **ICML MI Workshop**
- **Accepted** to the **ICML Mechanistic Interpretability Workshop** (Seoul); co-first author [\[paper\]](#)
 - Proposed and validated a framework showing SAE decoder-space geometry (neighbor density, max cosine similarity) predicts feature steerability before any intervention, achieving Spearman ρ up to **-0.546** ($p < 10^{-6}$) and AUROC up to **0.822**
 - Replicated geometry-steerability signal across 4 model architectures (Gemma-2 2B/9B, Llama-3.1-8B, Qwen3-8B) and 2 SAE widths (16K/65K); identified depth-boundary conditions where signal degrades
 - Designed end-to-end evaluation pipeline covering 100+ features $\times$ 100 prompts $\times$ 9 steering coefficients with coherence filtering, SALADbench refusal scoring, and permutation null tests
 - Optimized batched inference to reduce model calls by 94% (69K $\rightarrow$ 4.3K), cutting GPU compute cost on Lambda Cloud
- Auditing Behavioral Probes for Contextual Privacy** June 2026 – Present
- Built a mechanistic interpretability pipeline across 270 ConfAide privacy scenarios to test whether Qwen2.5-7B's internal activations predict upcoming disclosure behavior; compared linear probes against scenario-text baselines
 - Developed an evidence-span annotation rubric separating disclosure outcomes from response strategies; found 28/30 audited "refused" responses disclosed protected information, exposing a mismatch between judge labels and privacy protection
 - Identified a pre-generation signal for disclosure-limiting strategies such as hedging and redirection, achieving 0.728 AUROC (+0.162 over the strongest tested text baseline); verified robustness through nested layer selection and an evaluation-awareness control
 - Tested the signal's temporal and cross-model limits: a preregistered analysis found no strengthening in the eight tokens before disclosure, while an exploratory Llama-3.1-8B evaluation detected no comparable advantage over text baselines
- Mechanistic Interpretability for AI Financial Advisors (Mechanistic KYC)** May 2026 – Present
- Built and validated a rubric-based synthetic client profiler grounded in the Grable & Lytton risk-tolerance scale, generating 6,000 stratified client profiles with full unit-test coverage
 - Designing linear probes and activation patching on frozen Gemma-2-9B/Llama-3.1-8B to distinguish belief capture from hypocritical compliance under client pressure, including a novel self-vs-client risk direction decomposition

EXPERIENCE

- Technical AI Safety Course** June 2026 - Current
- BlueDot Impact* *Remote*
- Selected for BlueDot Impact's Technical AI safety cohort; studying current alignment techniques, interpretability, and model evaluations through weekly expert-facilitated discussions
- Research Assistant** January 2025 – June 2025
- Humanitarian Engineering and Social Entrepreneurship – Hexarech* *State College, PA & Kisumu, Kenya*
- Analyzed sensor data to adjust feed schedules and mitigate poor water conditions, cutting fish mortality 30–50%
 - Traveled to Kenya to co-design low-cost demonstration cages with local stakeholders (BMUs, hatcheries, extension officers)